Cambridge (CIE) A Level Computer Science

Your notes

Circuit Switching, Packet Switching

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Packetising

What is packetising?

- Packetising is the process of **dividing a large message into smaller**, manageable units called **packets**
- Each packet can be **transmitted individually** across a network

Structure of a packet

- Each packet typically contains:
  - **Header** – contains control information such as:
    - **Source IP address** – who sent the packet
    - **Destination IP address** – where it's going
    - **Sequence number** – used to reassemble packets in the correct order
    - **Protocol** – e.g. TCP or UDP
    - **Packet length** – the size of the packet
    - **Checksum** – used for error checking
  - **Payload** – the actual data being sent
  - **Footer (trailer)** – may include extra error-checking or control information

Packet transmission & reassembly

- Once packetised and encapsulated with headers, **packets are transmitted independently**
- They may travel along **different routes** through the network
- At the destination, packets are **reassembled** into the original message using sequence numbers

Circuit & packet switching

What is packet switching?

- Packet switching is a communication method where **data is broken into packets** and sent across a shared network
- Each packet may **take a different path to the destination** and is reassembled on arrival

Benefits:

- **Efficient** use of network **bandwidth**

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- More **reliable** – only lost or corrupted packets need to be resent
- **Lower cost** – network resources are shared

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Drawbacks:

- **Not ideal** for **real-time applications** like VoIP or video calls
- Packets can **arrive out of order** and require reassembly
- Network congestion may lead to **packet delay or loss**

What is circuit switching?

- Circuit switching is another communication method where **data is broken into packets** and sent across a shared network
- Circuit switching sets up a **dedicated communication path** between two devices for the duration of a session (like a traditional phone call)

Benefits:

- Ideal for **real-time communication** – steady and continuous data flow
- **No delays** once the circuit is established
- Data arrives **in order**

Drawbacks:

- **Less efficient** – resources remain reserved even when not in use
- **Higher cost** due to dedicated lines
- **Less scalable** and flexible for large networks

Packet switching vs circuit switching

Feature	Packet switching	Circuit switching
Definition	Data is split into packets and sent independently over shared routes	A dedicated path is set up for the duration of the communication
Transmission	Data is sent in individual packets	Data is sent in a continuous stream
Efficiency	High – uses bandwidth as needed, paths shared among many users	Low – resources are tied up for each communication, even when idle
Reliability	Robust – packets can be rerouted if a failure occurs	Less robust – if the path fails, the call is dropped and must be re-established

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Scalability	Highly scalable – handles large volumes of data and users	Less scalable – more users require more physical paths
Use Cases	Best for emails, web pages, file downloads	Best for real-time communication (voice calls, video conferencing)
Data Order	May arrive out of order – requires reassembly	Always arrives in order
Cost	Lower – due to shared infrastructure	Higher – due to reserved bandwidth and infrastructure

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Examiner Tips and Tricks

Try to avoid talking about the speed of data transmission in an answer to a question on packet or circuit switching. This will not get you a mark in the exam and, in some questions, is explicitly stated as not worthy of a mark. It is better to talk about higher bit rates or **bandwidth** (the number of bits sent per second) or the efficiency of the transmission

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